

Simplify the following expressions.

1. $\frac{3x^2 - 12}{x^2 - 4x + 4}$

2. $\log\left(\frac{100}{x}\right) - \log(10)$

4. $2^x \cdot 4^{x+1}$

5. $\frac{x^4 - 16}{x^2 - 4}$

$$6. \quad \frac{\frac{2}{x} + \frac{3}{y}}{\frac{1}{x} - \frac{1}{y}}$$

$$7. \quad \log_3(81)$$

$$8. \quad \frac{3^{2x} \cdot 9^x}{27^x}$$

$$9. \quad \frac{(x^2 - 9x + 20)(5^{x+2})(3x - 6)}{(x - 4)(x - 2)(25)}$$

10. $5^{2 \log_5(7)}$

11. $\frac{\log 216}{\log 6}$

12. Is this equation necessarily true, assuming that $a, b > 0$?:

$$\cos \left(\cos^{-1} \left(\frac{a}{\sqrt{a^2 + b^2}} \right) \right) = \frac{a}{\sqrt{a^2 + b^2}}$$